Visual Haystack

Abstract

Visual Haystack, p.1

> However, the ability to process a large number of visual tokens does not guarantee effective retrieval and reasoning for multi-image question answering (MIQA), especially in real-world applications like photo album searches or satellite imagery analysis

No guarantee of effective retrieval and reasoning for visual tokens even after long context

Visual Haystack, p.1

> e address these limitations by introducing a new vision-centric, long-context benchmark, “Visual Haystacks (VHs)”. We comprehensively evaluate both open-source and proprietary models on VHs, and demonstrate that these models struggle when reasoning across potentially unrelated images, perform poorly on cross-image reasoning, as well as exhibit biases based on the placement of key information within the context window

1. Open source and proprietary models
2. potentially unrelated images and poor performance on cross-image reasoning
3. Biases regarding position of information

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> lightweight visual-RAG framework that processes up to 10k images

Solution given - external RAG system, not a modification to the mechanism.

Conclusion

Visual Haystack, p.10

> With VHs, we then identified three key limitations across both open-source and proprietary models: susceptibility to visual distractors, difficulty reasoning across multiple images, and biases due to the positional placement of key information within the context window.

Three key limitations were observed

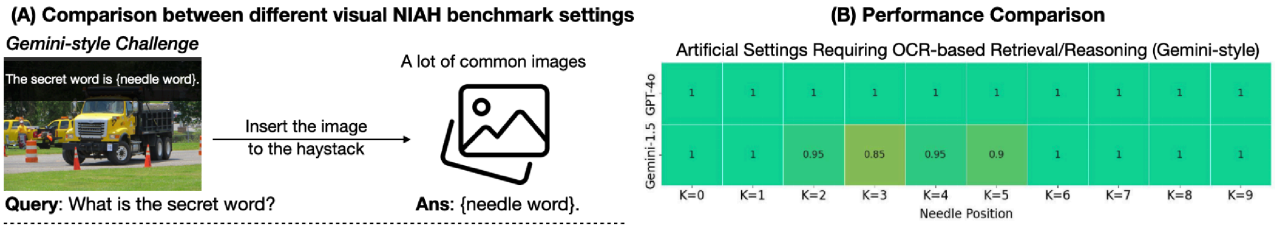
1. Susceptibility to visual distractors
2. Difficulty reasoning across multiple images
3. Biases due to positional placement of keys

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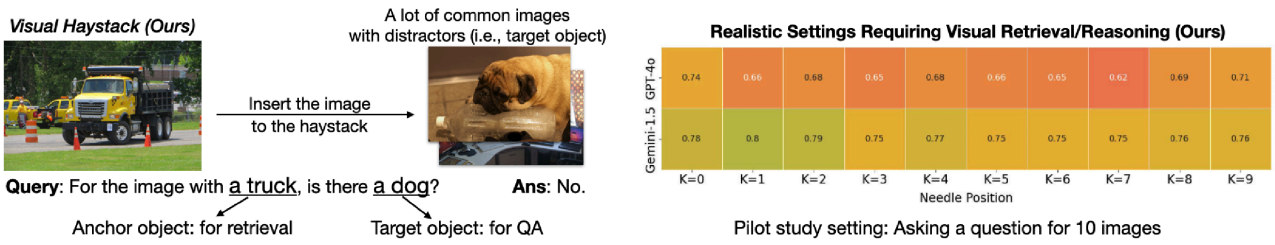
> dditionally, we found that contemporary LMMs are limited to processing no more than 100 images simultaneously. To address these limitations, we introduced MIRAGE, an open-source visual-RAG framework capable of handling up to 10,000 images on a single 40GB A100 GP

Authors claim that contemporary LMMs are not capable of processing 100 images simultaneously, however, our experiment shows failure in about 64 images for multihop reasoning.

Tables and figures

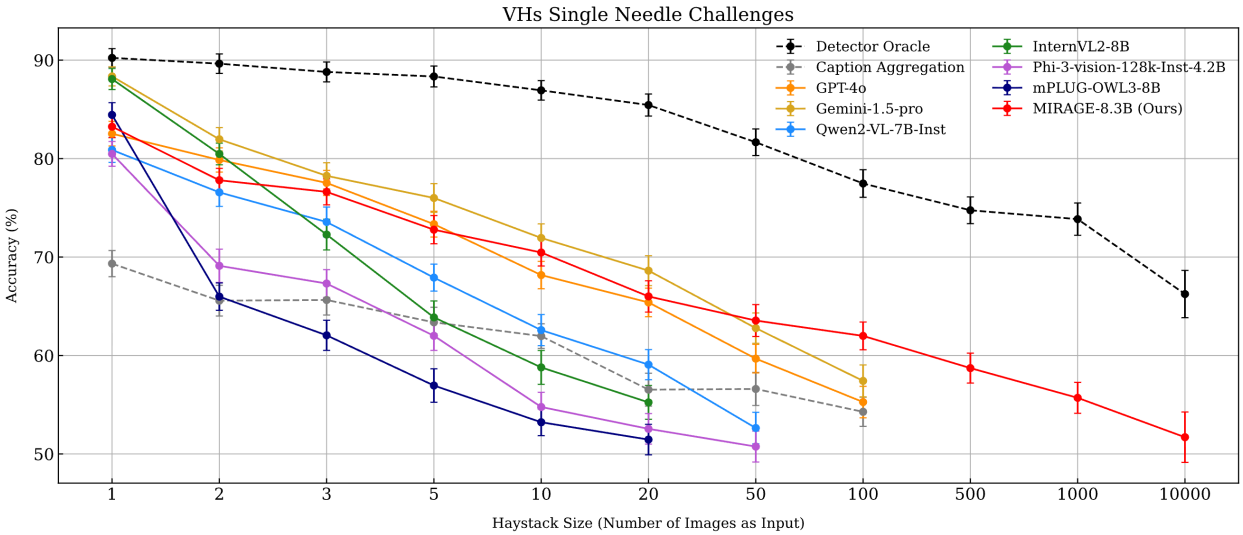


Traditional NIAH benchmarks add text containing the needle to an image. LMMs use their OCR capabilities to find the needle and achieve good performance.



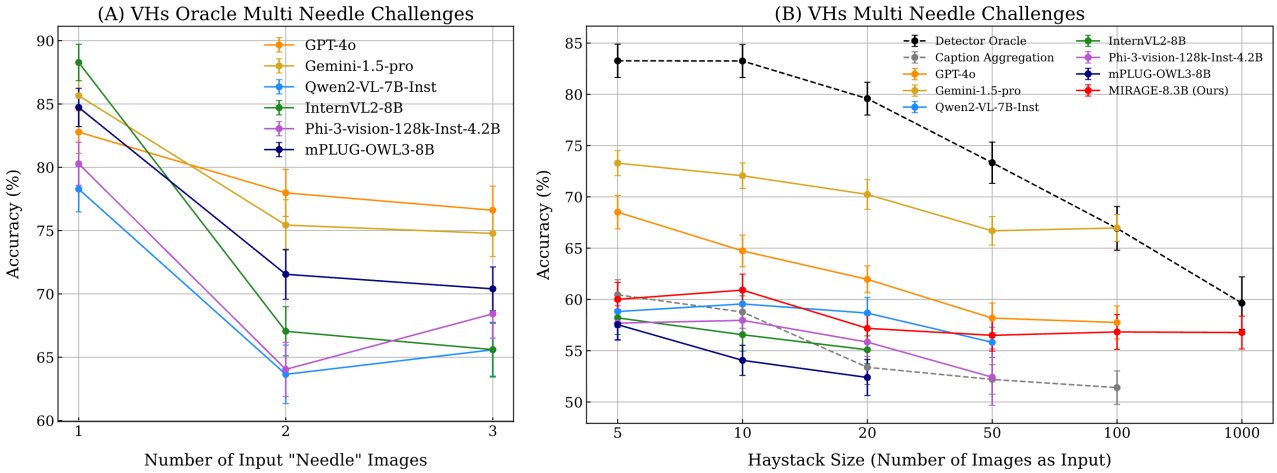
V-NIAH forces the model to find the correct image and then answer the question based on that image. Distractors contain the target, but without the question class.

1. Mixed classes are used (truck with dog)
2. Single hop retrieval

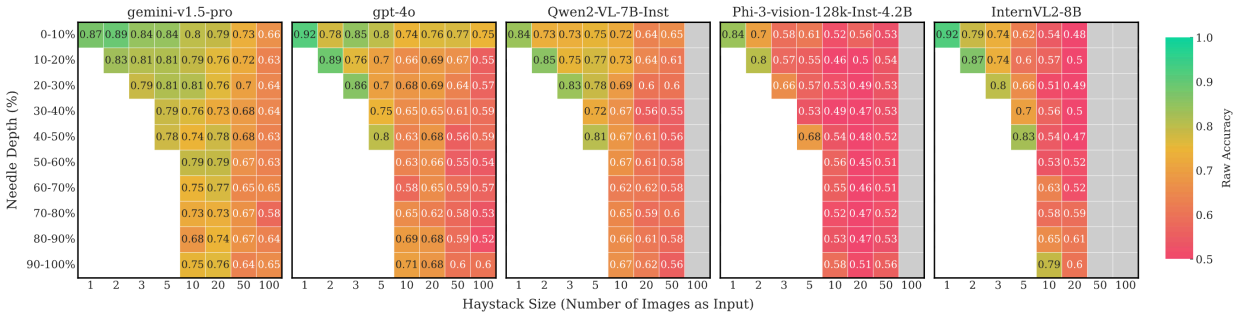


Some observations:

1. Number of images for single images is very large. Maybe the visual tokens exceed context window? 10k images is the maximum.
2. Accuracy range between 50-90% for single needle setting
3. General downward trend in accuracy
4. "Yes" and "no" are the only answer types.

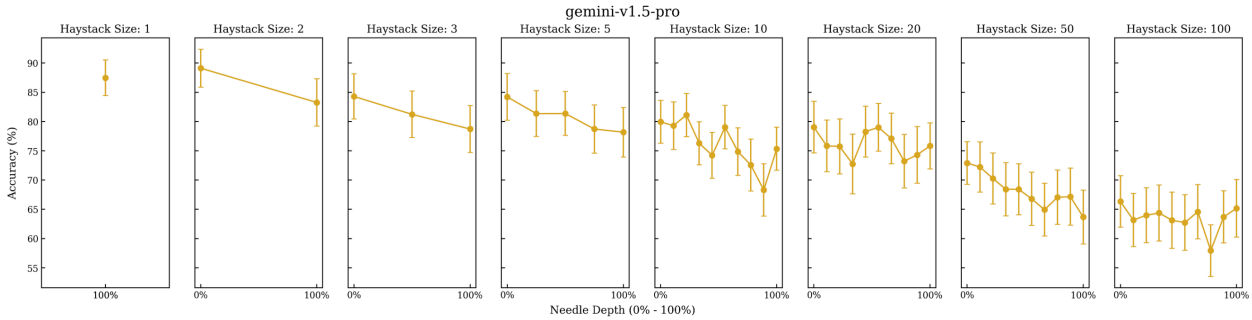


1. Experiment with 0 distractors - oracle experiment
2. Similar downward trend for multineedle challenge.
3. 7-B scale used (but solution proposed)



This graph indicates the position of the needle in [1, 2, 3, 5, 10, 20, 50, 100] distractors. Similar to this, we can also divide the distractor in not just 3 but more than 3 bins for better/fine-grained analysis.

NOTE: Authors report standard deviation and mean of the scores (We would have to do the same for at least three seeds)



Introduction

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> However, fitting more visual tokens within a context window does not inherently translate to improved performance in more complex, multi-image question-answering (MIQA) scenarios. In fact, we demonstrate in Sections 2.1 and 3 that long-context LMMs are unable to accurately retrieve or reason across thousands of images,

More context length is not a guarantee of good reasoning and strong retrieval.

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> LMMs must integrate and reason about visual information spread across multiple images to perform well on VHs.

LMMs must actually look at the images (a test of their visual grounding) before answering the question. The target is also part of the image, moving away from overlaying text in an image (thus separating the reliance on OCR from visual evidence)

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> While most LMMs can process up to 100 images, they exhibit up to a 40% performance drop on VHs compared to non-retrieval QA tasks, implying that LMMs struggle with retrieval and are prone to visual distractors.

Process upto 100 images, but performance drops as distractors increase.

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> his performance decline is even more severe when multiple images are used as needles, suggesting that LMMs struggle to reason across multiple images

Low multi-image reasoning

Visual Haystack, p.2

> e observed significant positional bias in LMMs, reminiscent of the “lost-in-the-middle” pathology found in NLP tasks (Liu et al., 2024b). Models’ sensitivity to the placement of key information within the context window often results in sub-optimal performance when the needle is positioned unfavorably.

Positional bias exists.

Result and discussion

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> he needles in VHs are real objects within natural images, overcoming the artificial inclusion of overlaid text or image patches (Reid et al., 2024; Wang et al., 2024c) that previous visual NIAH benchmarks suffer from.

Real objects within real images.

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> VHs is much more reliable. Finally, VH is diverse and large-scale; haystack sizes are as large as 10,000 images

Is context length not exceeded for 10k images?

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> With over 1,000 examples of each setting, VHs provides 97k images total for comprehensive visual NIAH evaluation.

Large scale of evaluation

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> The answers are binary (yes/no).

VH uses binary answers. The problem with binary answers is we do not know if the model actually looked at the right image.

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> fter selecting the object pairs, setting the questions, and determining the answers, we then compile corresponding image haystacks by first curating the needle images, which contain the “anchor object” as indicated by the paired answer

Benchmark construction details

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> However, some of them contain the target objects so as to create meaningful distractors.

Interesting experiment for us to see. We can also include images of target animals in out setting.

Visual Haystack, p.4

> Our analysis yielded three key insights into LMM behavior on long-context visual retrieval and reasoning tasks: (1) a susceptibility to visual distractors, (2) difficulty in reasoning across multiple images, and (3) a bias in relative image positioning.

Summary of findings

Visual Haystack, p.4

> As shown in Figure 2, the single-needle challenge reveals that LMMs are significantly impacted by visual distractors. When presented with only one image, general-purpose LMMs such as InternVL2-8B and Gemini v1.5 Pro perform comparably to specialized detectors like OWLv2, indicating that these models can handle simple visual reasoning tasks in isolation. However, as the number of input images increases, a notable drop in performance is observed compared to an oracle baseline that combines a detector with a language parser. This degradation, which does not appear in prior artificial OCR-based NIAH benchmarks as shown in Figure 1 (B), suggests that the main limitation of current LMMs lies in their ability to retrieve relevant visual information from large sets of images containing distractors

Single image performance is high, while performance degrades as the number of images increase. This points to: multi-image lmm's actually behave well with single images.

Visual Haystack, p.5

> Interestingly, we observed a sharp performance decline across all LMMs as the number of needle images increased from one to two (Figure 3 (A)). However, as the number of needle images increased from two to three, the performance decline became much smoother or even stabilized

Performance drops as the number of needle images increases from 1 --> 2, then mostly stabilizes.

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> As shown in Figure 4, LMMs are also sensitive to the positional placement of the needle image. Interestingly, this positional bias varies across models. For example, Gemini v1.5 Pro tends to favor images at the beginning of the sequence,

> while GPT-4o exhibits a clear lost-in-the-middle effect, similar to what has been observed in natural language tasks (Liu et al., 2024b). In contrast, open-source models like Qwen2-VL, Phi-3-vision, and InternVL2 favor images closest to the question (i.e., the last image in the sequence) given a small number of images (N ≤5

Positional bias exists but trends vary for different models.

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> While its simplicity and unprecedented scale are key strengths, the primary limitation of the benchmark is the scope: since it is based on MS-COCO images, it inherits the implicit biases in the dataset including notable gender, race, and location biases (Hendricks et al., 2018; Bhargava & Forsyth, 2019; Hirota et al., 2022; Wang et al., 2022). It is important to work in the future toward developing benchmarks that do not favor models that prefer data having such biases. Beyond such implicit biases, COCO objects are also limited to 80 categories - strong performance on the VHs benchmark does not imply that the model will generalize well to all MIQA problems.

Future scope:

1. A benchmark without gender, race and location biases.
2. COCO-based benchmarks are limited to 80 categories